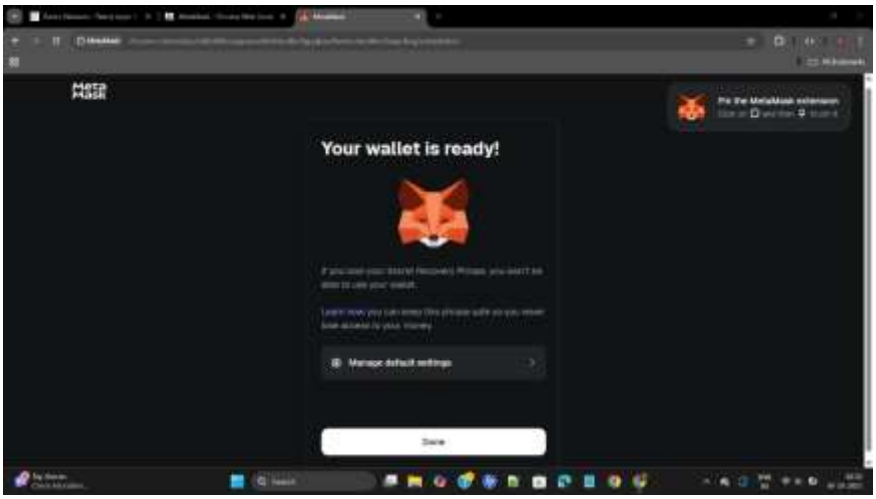
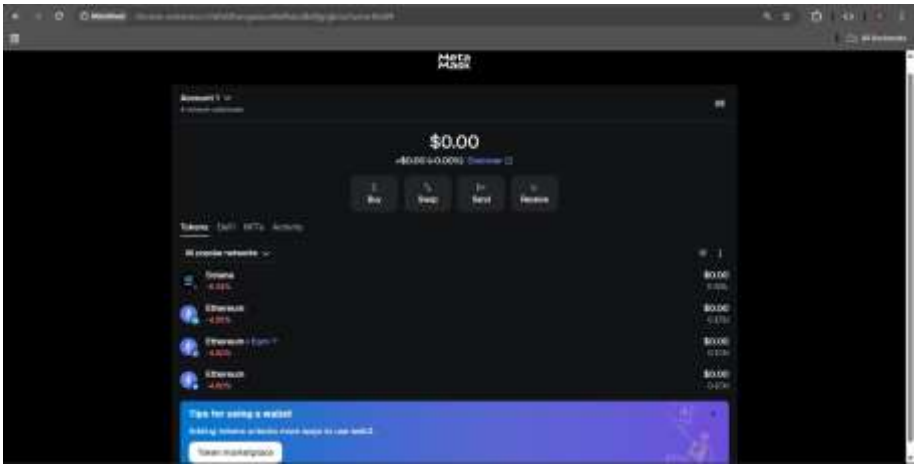
Web3 and Blockchain Basics – Documentation Report

1. MetaMask Wallet Setup

To start my Web3 journey, I installed the MetaMask browser extension from its official website (https://metamask.io). After installation, I created a new wallet account, configured a strong password, and securely stored the generated seed phrase offline to maintain wallet safety. MetaMask offers a user-friendly interface that allows users to manage digital assets efficiently and seamlessly connect with various decentralized applications (DApps) on the blockchain network.

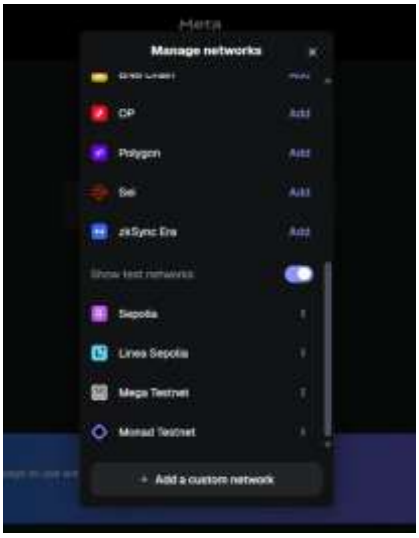


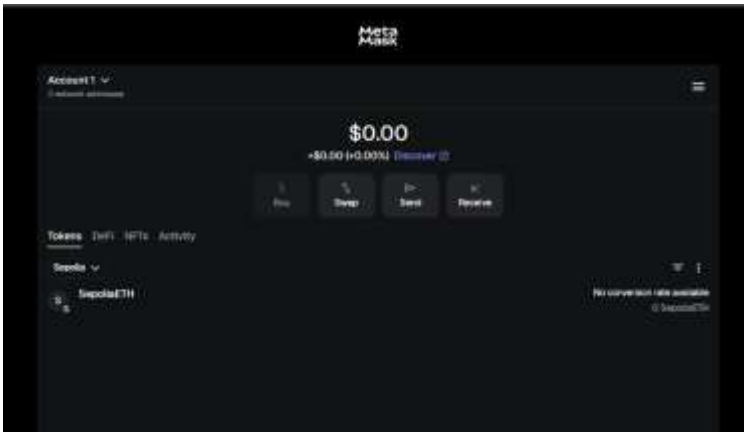
After creating the wallet, I accessed the home screen showing my wallet address and ETH balance.



2. Testnet Configuration

After setting up the wallet, I enabled the **Show Test Networks** option in the MetaMask settings to gain access to the Sepolia Test Network. In the updated MetaMask interface, test networks such as Sepolia can be added directly through reliable network providers without requiring manual configuration. I connected my wallet to the Sepolia Test Network using a trusted source to ensure a secure and seamless setup.





Although I couldn't obtain Sepolia ETH from the available faucets (as most required a small balance on the mainnet), I successfully learned how to configure MetaMask for blockchain test network usage and understood the process of connecting decentralized wallets to test environments.

3. DApp Interaction

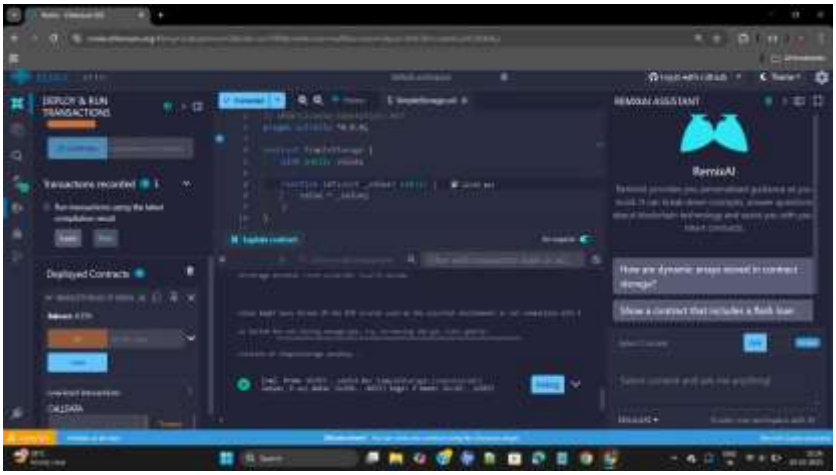
(a) Remix IDE – Smart Contract Deployment

I utilized the Remix Ethereum IDE to develop and deploy a basic Solidity smart contract named *SimpleStorage*. This contract is designed to store and retrieve a numerical value on the blockchain.

The execution environment was configured to **Remix VM (Shanghai)**, which acts as a local blockchain simulation and does not require real ETH or an active internet connection.

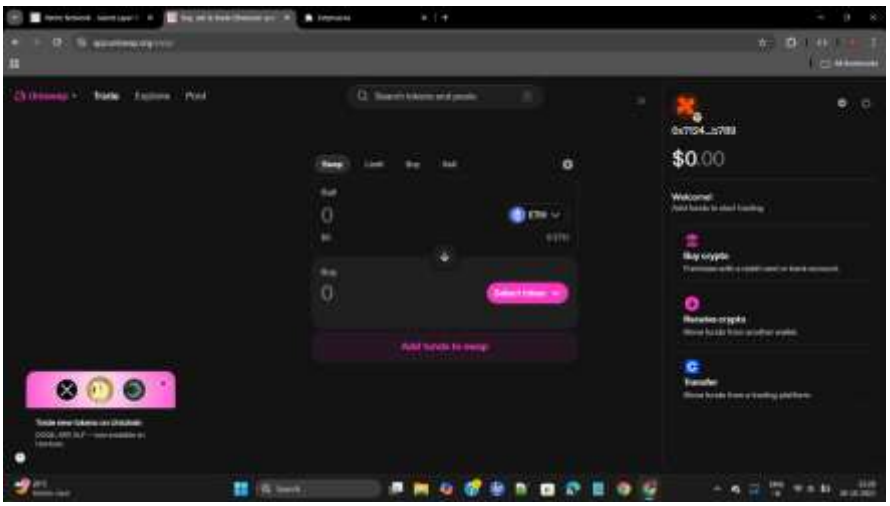
I carried out the following operations:
• store(15) → Saved a numeric value on the blockchain
• retrieve() → Fetched the stored value

This process demonstrated a typical DApp interaction scenario, where users interact with smart contracts by sending transactions and reading data from the blockchain.



(b) Uniswap DApp Exploration

Next, I accessed the Uniswap decentralized exchange platform and successfully connected my MetaMask wallet. Due to insufficient Sepolia ETH, I was unable to execute token swaps; however, I explored features such as gas fee estimation, wallet connection requests, and transaction previews. This provided practical understanding of how real-world DApps handle user transactions and interactions on the blockchain network.



4. Written Reflection

My exploration of Web3 technology enabled me to understand the fundamental differences between blockchain systems and traditional databases. In centralized databases, data is managed and controlled by a single authority or organization, whereas blockchain stores information on a distributed ledger maintained across multiple nodes in the network. Every transaction is cryptographically secured and verified using consensus mechanisms such as Proof of Work (PoW) and Proof of Stake (PoS), which ensure transparency, security, and immutability of data.

One of the key innovations in blockchain technology is the concept of smart contracts — self-executing programs that automatically perform predefined actions when certain conditions are fulfilled. By working with the SimpleStorage contract, I gained practical experience in understanding how decentralized applications (DApps) utilize these contracts to deliver automated and trust less services without the involvement of intermediaries.

Using MetaMask further improved my awareness of wallet security and transaction authorization. The seed phrase plays a crucial role in protecting wallet access, as anyone possessing it can gain complete control over the associated assets. Therefore, I ensured that it was stored securely in an offline environment and took necessary precautions such as verifying website URLs and avoiding phishing attempts.

One of the primary challenges I encountered was obtaining testnet ETH. Several faucets required users to have a minimum mainnet balance, which initially limited my ability to perform transactions on the Sepolia network. To overcome this issue, I used Remix VM, a simulated blockchain environment that allowed me to deploy and interact with smart contracts without needing actual tokens.

Additionally, I observed that blockchain transactions involve gas fees, which represent the computational cost required to execute operations on the network. These fees vary depending on network congestion and the complexity of the transaction. Deploying and interacting with smart contracts also helped me understand transaction confirmation times and gas estimation mechanisms.

Overall, this experience provided me with a solid foundation in blockchain technology, covering essential aspects such as wallet setup, security practices, and decentralized application interaction. It served as a comprehensive introduction to Web3 concepts, highlighting the principles of decentralization, transparency, and user ownership.

5. Technical Summary

Field	Details
Testnet Used	Sepolia
DApps Interacted With	Remix IDE and Uniswap
Transactions Performed	Local contract deployment, store & retrieve operations
Wallet Used	MetaMask browser extension
Transaction Type	Simulated transactions (Remix VM)
Errors/Challenges	Could not obtain Sepolia ETH due to faucet restrictions
Troubleshooting Steps	Used Remix VM (Shanghai) environment
Security Practices	Seed phrase stored ofline, URLs verified, avoided phishing

6. Conclusion

This task offered valuable hands-on exposure to the core concepts of Web3 and blockchain technology. Through this activity, I gained knowledge about distributed ledger systems, consensus mechanisms, smart contracts, and the overall architecture of decentralized applications (DApps). I was able to configure a digital wallet, explore test network settings, deploy a smart contract using Remix, and connect my wallet to Uniswap, all while following essential security best practices.

Although I initially faced difficulties in acquiring testnet ETH, the use of Remix VM enabled me to safely simulate DApp interactions without requiring real tokens. This experience enhanced my confidence in working with blockchain tools and provided a practical understanding of the decentralized ecosystem in a secure and risk-free environment.